ASSIGNMENT 24

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Domain:data science

Since RESULT is missing, we define:

Pass (1) if Study Hours ≥ 5 AND Attendance ≥ 60 Otherwise → Fail (0)

Student	Study Hours	Attendance	Result
A	2	40	Fail (0)
B	3	50	Fail (0)
C	5	60	Pass (1)
D	6	70	Pass (1)
E	7	80	Pass (1)
F	4	65.5	Fail (0)
G	6	70	Pass (1)
H	8	33	Fail (0)
I	10	80	Pass (1)
J	7	25	Fail (0)

a. NAIVE BAYES

Convert into categories:

Study Hours: Low (<5),High (≥5)

Attendance: Low (<60),High (≥60)

Count values:

Pass (1): C, D, E, G, I → 5 students

Study High = 5

Attendance High = 5

P(Study=High | Pass) = 5/5 = 1

P(Attendance=High | Pass) = 5/5 = 1

Fail (0):A, B, F, H, J → 5 students

Study Low = 3 (A, B, F)

Study High = 2 (H, J)

Attendance Low = 4

Attendance High = 1

P(Study=Low | Fail) = 3/5

P(Attendance=Low | Fail) = 4/5

Based on probability calculation → Students with high study hours and attendance are classified as PASS.

b. GAUSSIAN NAIVE BAYES

Now calculate Mean (u)

For PASS students:(5,6,7,6,10)

Mean Study: u=(5+6+7+6+10)/5=6.8

Attendance:

(60,70,80,70,80)

u=72

For FAIL students:(2,3,4,8,7)

Mean Study: u=4.8

Attendance:

(40,50,65.5,33,25)

u=42.7

Based on mean values → Students closer to (6.8, 72) are classified as PASS.

c.SVM

Student C:

5 + 60 = 65 → PASS

Student A:

2 + 40 = 42 → FAIL

Using separating line → Students above boundary line are PASS, others are FAIL.